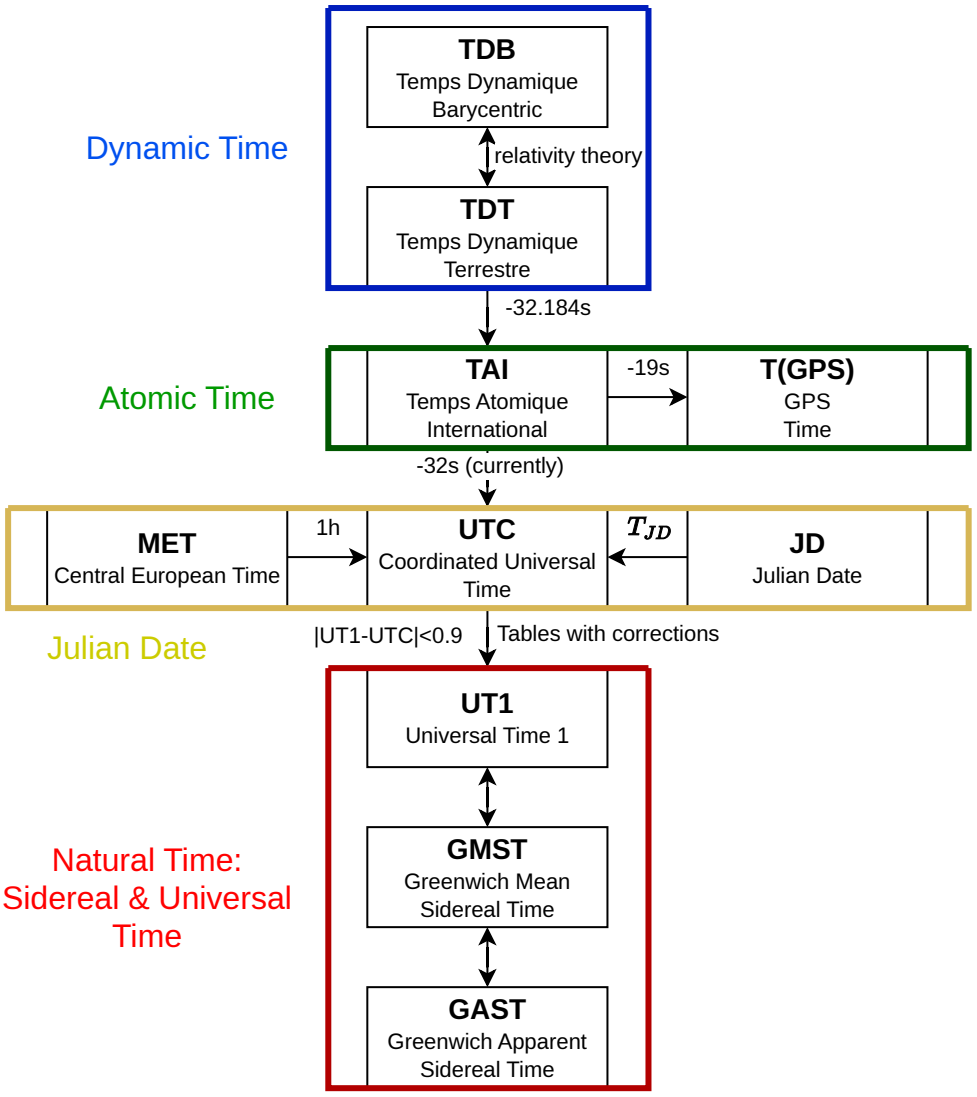
Lecture 6 & 7: Principles of timing systems and application in Earth observation system

Time Systems

- Dynamic Time System
  - Based on orbital motion of celestial bodies in solar system
  - Useful for describing satellite motion
- Atomic Time System
  - Based on processes in nuclear physics
  - Useful for positioning, relativity, astronomy, etc.
- Civilian Time System
  - Namely: Julian Date
  - Based on number of days elapsed since Jan 1 4,713 BC.
  - Useful for scientific computations
  - UTC for connection of all 3 time systems
- Natural Time System
  - Namely: Sidereal Time and Universal Time
    - Defined based on diurnal rotation of Earth
  - From Earth motion to atoms oscillation
  - useful for **Earth-based observations** in the CTRFs

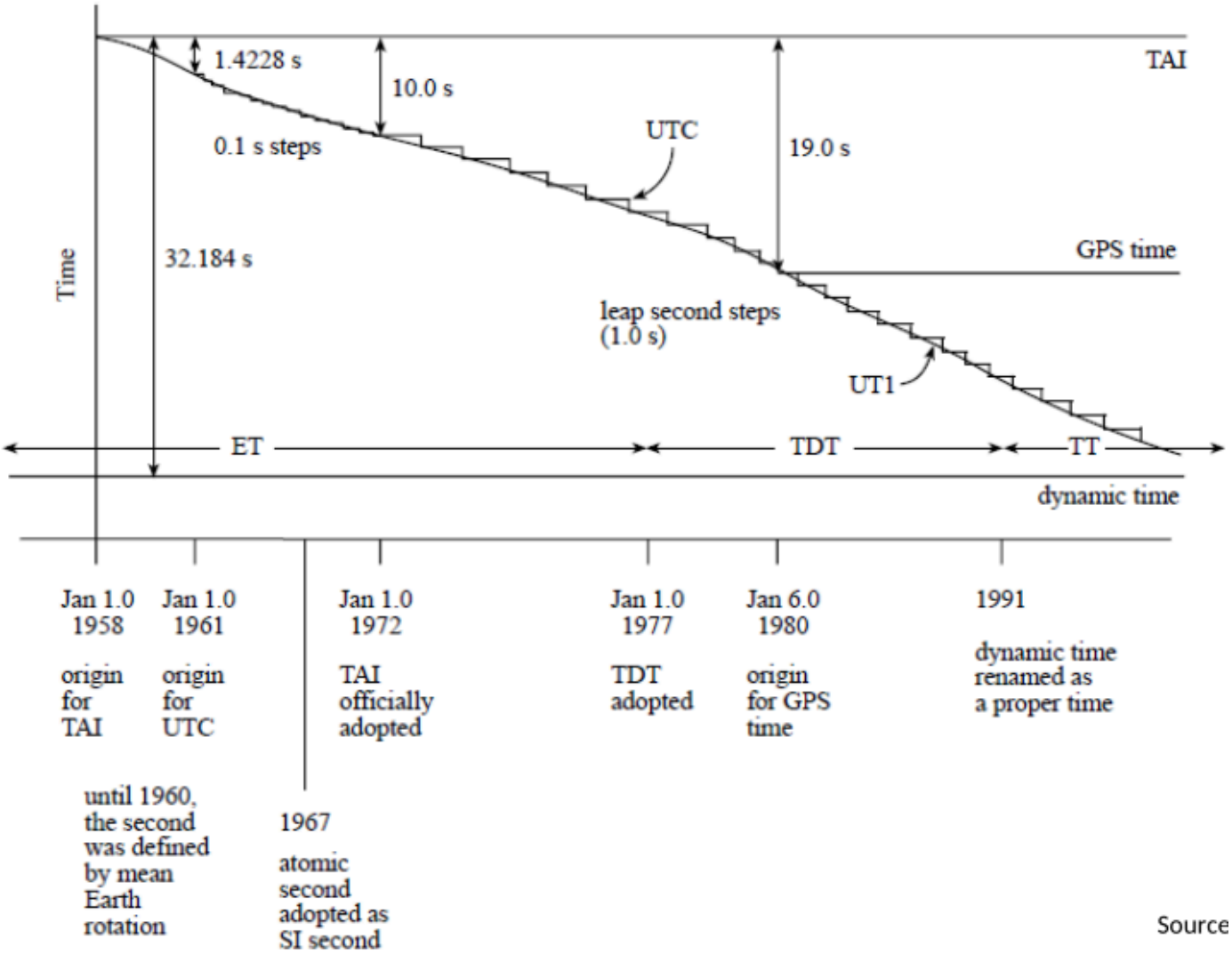
- for transferring between CCRF to CTRFs



Requirement for Time System Definition

1. **Periodic Event**
   - Rotation of Earth, orbital motion of Earth around sun, orbital motion of moons around Earth, Pendulum motion, motion of atoms, etc.
2. **Counter**
3. **Reference Epoch and Time Scale**
   - Amount of events within specific interval

Relationship between atomic scale and dynamic time scale

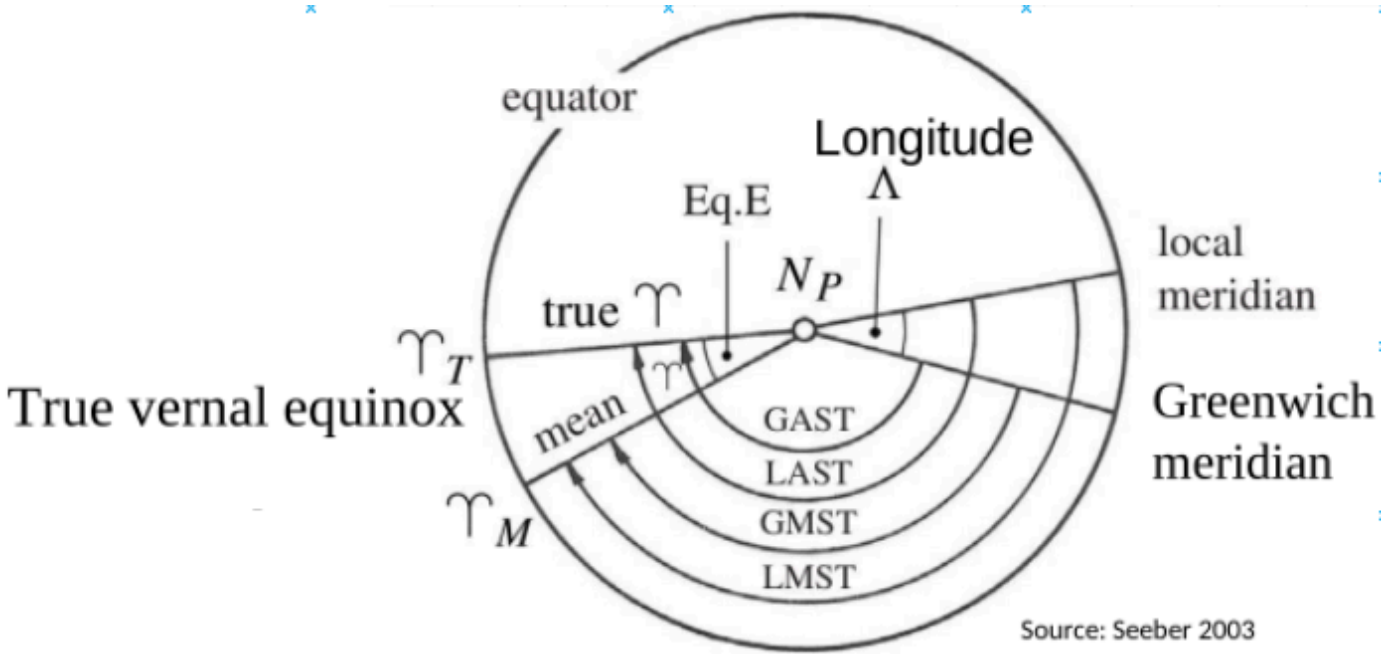


Source

Definition of Second

- Until 50s: Second based on length of mean solar day
  - Mean solar day = 24h*3600=86400s
  - 1s = 1/86400 mean solar day
- But: Earth's rotation not uniform enough to measure second: Short-term & Long-term irregularity
- -> Redefine second on **orbital (Newtonian) motion of Earth around sun**
  - Solar (tropical) year (Based on sun's ephemeris)
  - **Ephemeris second** (adopted 1952) as fraction of length of **solar day** at 1900
- Not considered: Relativistic Theory & Effects of planet etc. on Earth rotation around sun
- -> Dynamic time scale

Sidereal Time

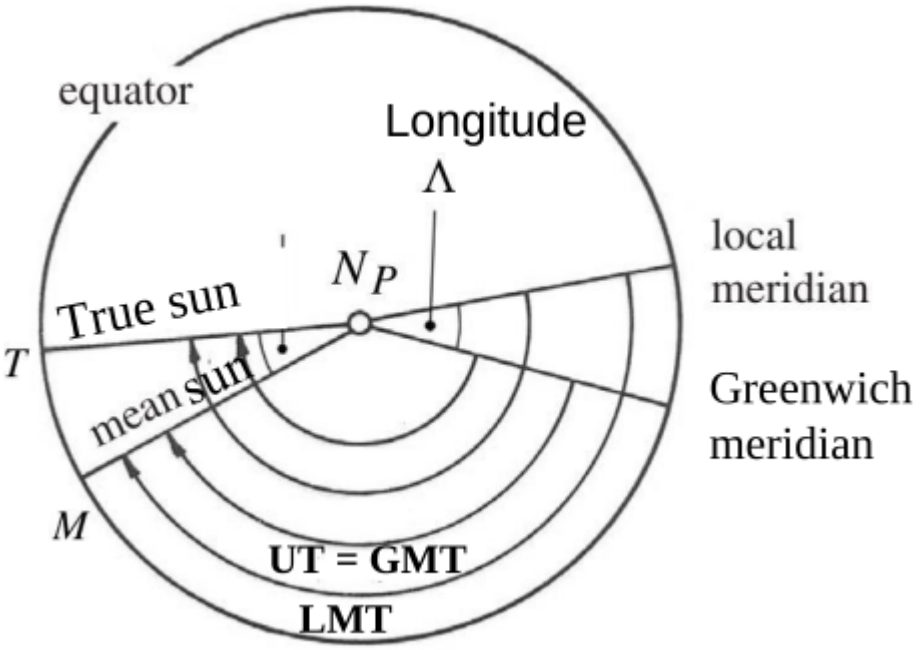


- Definition: hour angle of vernal equinox
  - Defined in quasi inertial CCRS
  - reflects actual rotation of Earth & precession and nutation effects of equinox
- Difference between **Apparent** ST (AST) and **Mean** ST (MST) due to nutation
- Scale in ST system: **mean sidereal day**
  - 2 consecutive transits of the mean vernal point accross the same meridian (corrected for polar motion)
    - 1 sid. day = 24 sid. hrs = 86400 sid. sec
- Relationship between AST & MST alias **Equation of the Equinox**
  - $AST = MST + \Delta\Psi \cos\varepsilon$
    - $\Delta\Psi$ is nutation component in longitude
    - ε is inclination angle
- Attention: **Precession**: Mean Vernal Point move w.r.t true vernal point
  - m = 4612.4362 (arcSec/cent.) + 2.79312 (arcSec/cent^2)T
    - T is Julian centuries
  - 24 hours of mean ST less than one complete rotation
    - ST day 0.0084sec shorter
- **Remember**: GAST for transformation from CCRS to **mean** CTRS

Universal Time UT

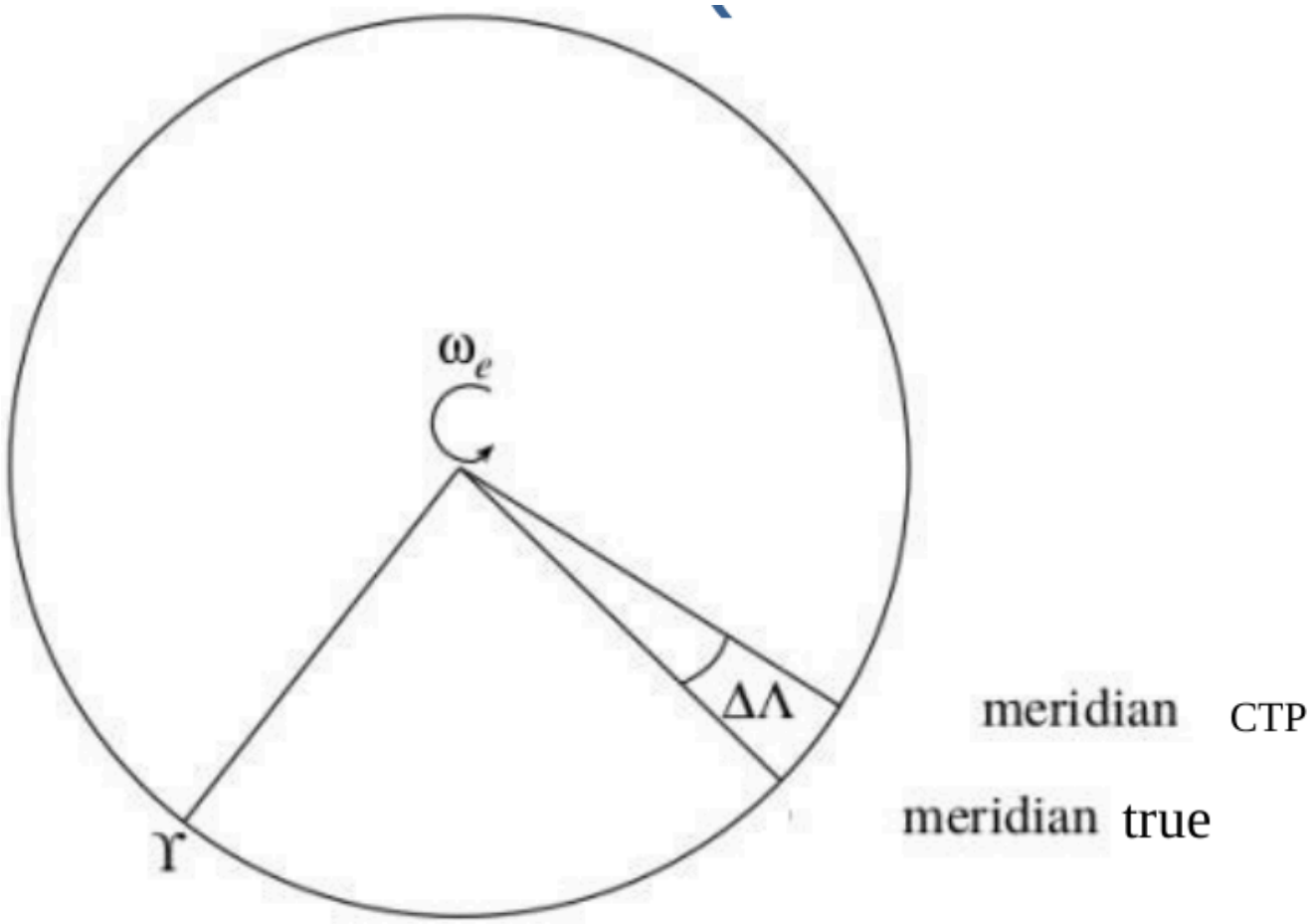
- Definition: Hour Angle of Sun
  - In **CTRS**

- Actual rotation of Earth and effects of polar motion
- No Precession error, like vernal point
- Fictious mean sun moving on equator with constant velocity
- Scale
- : **Mean Solar day**: 2 consecutive transits of the mean sun across same meridian (corrected for polar motion)
- 1 Sol. day = 24 Sol. hrs = 86400 Sol. sec



Irregularity in UT

- Irregularity due to Earth's rotation in CCTS
- UT from observation at station
  - local meridian w.r.t instantaneous pole
  - rotation axis varies due to polar motion
- **UT0**: UT from observations w.r.t true (instantaneous) pole position
  - contains variation in rotation due to polar motion
- **UT1**: determined w.r.t CTP
  - corrected polar motion
- $UT1 = UT0 + \Delta\lambda_{pole}$

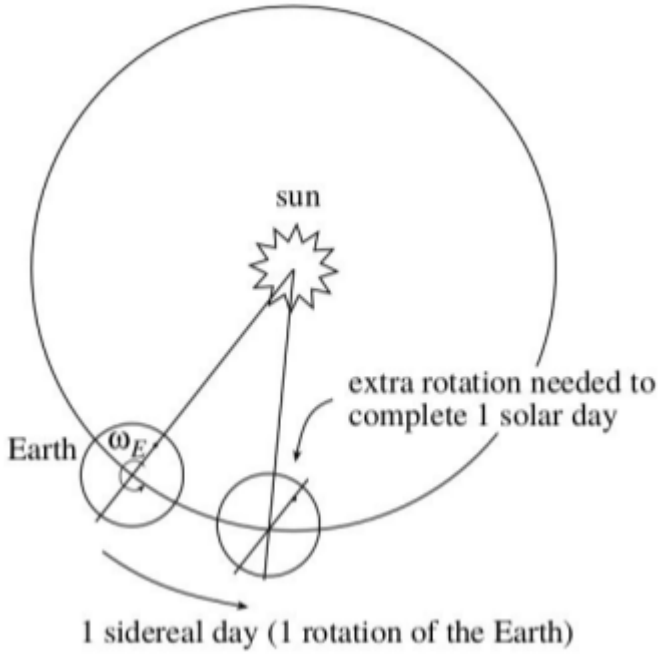


-
- UT1 still experiences small irregularities

Relationship between UT and ST

- Mean solar day > Sidereal day
  - Due to orbital motion of Earth around sun
  - Sun in different position on celestial sphere for next crossover

1 mean solar day = 24h 03m 56.5554s in sidereal time
1 mean sidereal day = 23h 56m 04.0905s in solar time
1 mean sidereal day = 1 mean solar day – 3m 55.909s



Julian Date

- In Scientific work: Continuous counting/dating more convenient
  - Civil Calendar has months and years of varying length
- 36 525 days per century
- Fundamental Epoch: 12h UT on 01.01.4713 B.C.
  - sufficiently preceding historical record
- Julian day number as day in continuous count since fundamental epoch
- JD of **standard epoch** of UT: J2000.0
  - 2 451 545.0JD = 12H UT on 01.01.2000
- For convenience: **Modified Julian Date (MJD)**:
  - MJD = JD - 2,400,000.5
  - Subtracted half a day, to start at midnight

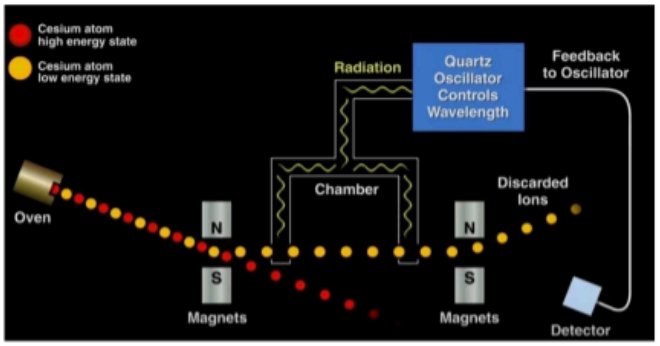
Atomic Time

- First realised in 1955
- **Origin** for atomic time officially: **UT1 time 0h 0m 0s (midnight) 01.01.1958**
- International Atomic Time (TAI) officially adopted January 1972
- TAI realised over 200 high-precision atomic clocks around the world located at sea level
- Worldwide synchronisation about 100nanoSec
- Alias: **Statistical clock**/time
  - TAI computed (like weighted average) from many atomic clocks

- All civil clocks set w.r.t. a TAI standard

Concept

- Scale scaled to the duration of the ephemeris second
- Precision: Consider **General Relativistic** effects from spatially varying gravitational potential
  - -> defined for **rotating geoid** (mean sea level) in Earth frame not in barycentric system
- Atomic Second: Duration of 9 192 631 700 period of the radiation
  - Corresponding to transition between 2 hyperfine levels of ground state of Cesium 133 atom
  - e^- close to nucleus have higher energy
  - Known energy gap between low and high energy
  - Apply exactly the specific amount of energy via microwaves ($9.192631700 MHz$)



1. Cesium atoms are moved down inside a tube where they pass through microwave field. If the generated microwave frequency is exactly 9 192 631 770 cycles per second (an ideal frequency), then the cesium atoms change their energy state to higher state.
1. A detector at the end of the tube gives an number of cesium atoms reaching it (those have changed their energy states only reach the detector).
2. Peaks appears in number of outputs (detected high energy atoms) when the real microwave frequency is exactly to ideal frequency 9 192 631 770 Hz.
3. This peak number is then used to make the fine correction necessary to bring the oscillator control with microwave exactly on the ideal frequency.
4. The number of (locked) corrected cycles (N) is then divided by the ideal frequency 9 192 631 770 to define the an atomic second.

Note that the general relation between number of cycles (N) and a given time interval ($t-t_0$) is $(t-t_0) = N/F_I$ where F_I is the ideal frequency and is 9 192 631 770 Hz to generate time interval $(t-t_0) = 1$ sec.

Atomic Clock Errors

- Oscillator errors (Oscillator producing frequency) are main part
- $f_i(t) = \underset{\text{Ideal Frequency}}{f_I} + \underbrace{\Delta f_i + \dot{f}_i(t-t_0)}_{\text{Systematic Error}} + \underbrace{\tilde{f}_i(t)}_{\text{Random Error}}$
- Δf_i: constant frequency bias of the oscillator C_i
- $\dot{f}_i$: frequency drift
- $\tilde{f}_i$: random frequency error

Corresponding error in time:

- $\Delta t_i = T_i(t_0) + R_i(t - t_0) + \frac{D_i}{2}(t - t_0)^2 + \int_{t_0}^{t} y(t)dt$
  - $T_i(t_0)$: constant time bias
  - $R_i = \frac{\Delta f_i}{f_I}$: time drift
  - $D_i = \frac{\hat{f}_i}{2f_I}$: Quadratic term (drift rate, ageing)
  - $y(t) = \int_{t_0}^{t} \frac{\tilde{f}_i(t)}{f_I} dt$
- For each clock: Determine first 3 terms
  - E.g. estimated at receiver for GPS positioning
- Comparison of many different clocks around the world

Stability of the Clock

- Performance in sample variance of N differences of relative frequency error over specific time interval τ

$$\Delta t_i = T_i(t_0) + T_i(t - t_0) + \frac{D_i}{2}(t - t_0)^2 + \underline{\int_{t_0}^{t} y(t)dt}$$

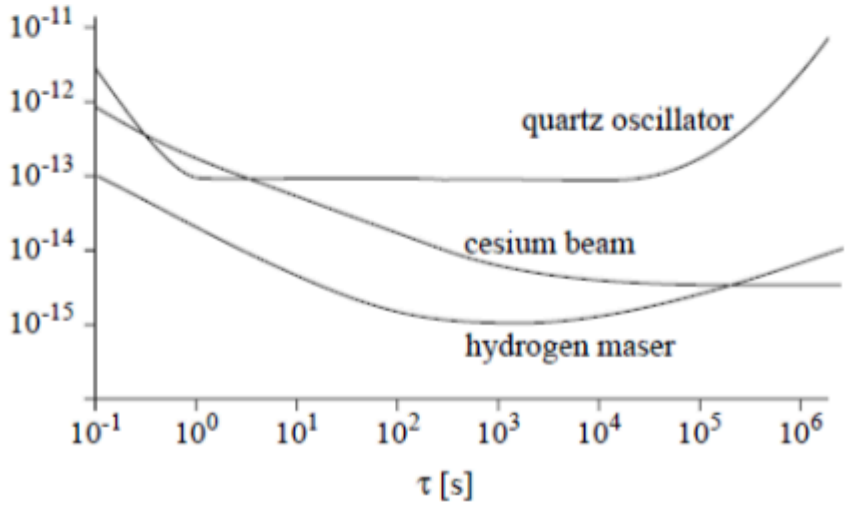
- Known as **Allen Variance**: Random Frequency stability of the oscillator

$$\sigma_y^2(\tau) = \frac{1}{N} \sum_{k=1}^{N} \frac{1}{2} (\bar{y}_{k+1} - \bar{y}_l)^2$$

- Atomic clocks show stability as function of τ
  - Allen variance decreasing, as τ increases from near zero to an interval of seconds
  - Allen variance reaches minimum over some range of τ
    - There it shows stability of the clock

Stability of Various Clocks

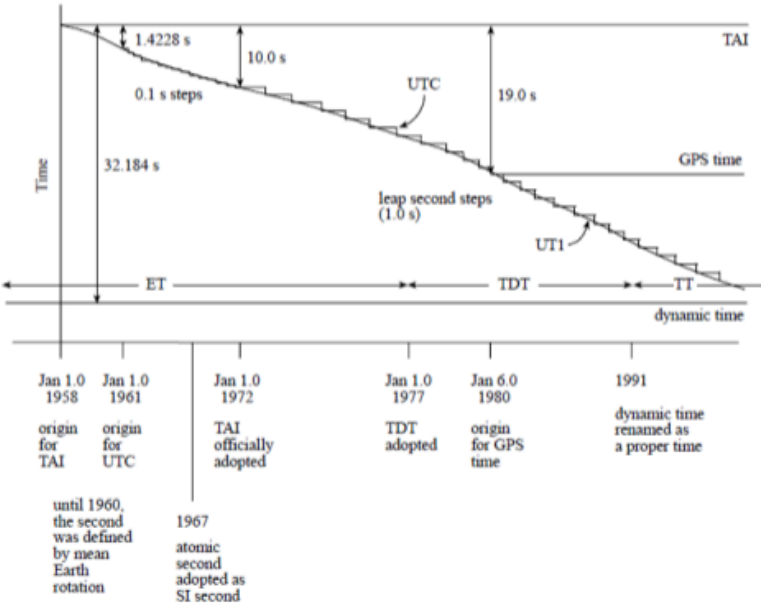
Clock		stability (min σ_y)	range of τ
quartz oscillator		$\geq 10^{-13}$	$1 \leq \tau \leq 1$ day
cesium beam,	laboratory	1.5×10^{-14}	several years
	commercial	2×10^{-12}	$\tau \leq 1$ year
		3×10^{-14}	$\tau \leq 1$ day
	Block II GPS	$O(10^{-14})$	$\tau \leq 1$ day
rubidium	laboratory	$\geq 10^{-13}$	$\tau \leq 1$ day
	GPS	2×10^{-13}	$\tau \leq 1$ day
hydrogen maser		2×10^{-15}	$10^3 \leq \tau \leq 10^4$ s



TAI and UT1

- Earth rotation **slowing down**
- Difference between UT1 and TAI increases

$$
\begin{aligned}
\mathrm{TAI} - \mathrm{UT1} &= +6.^{s}1 && \text{on January 1, 1968} \\
&= +16.^{s}4 && \text{on January 1, 1978} \\
&= +23.^{s}6 && \text{on January 1, 1988} \\
&= +30.^{s}8 && \text{on January 1, 1998} \\
&= +31.^{s}9 && \text{on January 1, 2001} \\
&= +32.^{s}3 && \text{on January 1, 2003.}
\end{aligned}
$$



UTC

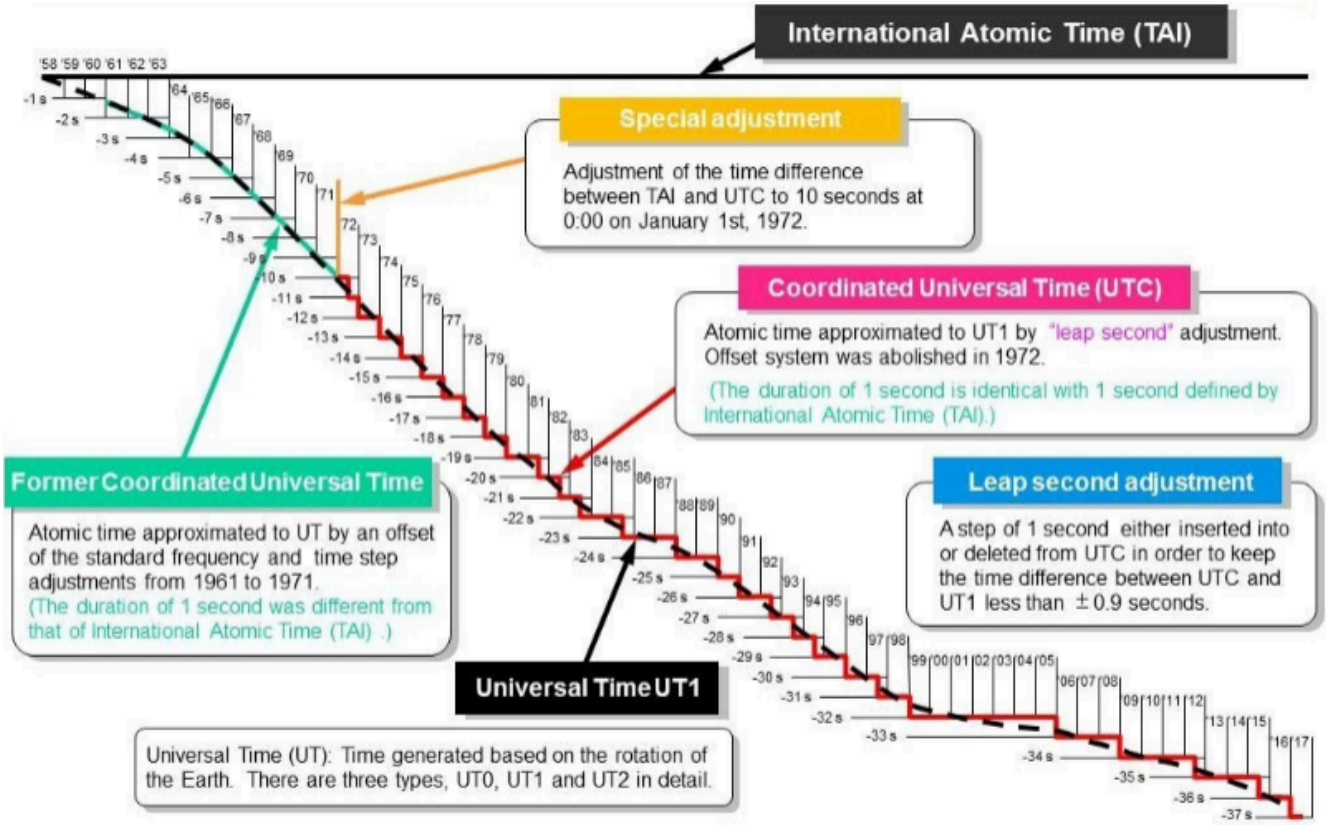
- TAI is highly uniform time scale
- But: Still need to correspond to UT1 (solar time)
  - New atomic time syntonised to TAI
  - Close to UT1
- Universal Time Coordinated UTC defined in 1972

$$|UT1 - UTC| < 0.9s$$

- UTC equal to TAI but different from TAI by integer number of second (**leap second**) as:

$$TAI - UTC = n \cdot /1s$$

- n - leap second



- DUT1=|UT1-UTC|< 0.9 sec and UTC is broadcast so that you can determine UT1.
- Current difference TAI and UT1 is 10 + 27 leap seconds = 37 seconds

GPS Time

- Atomic time consistent with TAI with $1\mu s$
- Zero epoch: UTC at 0h 06.01.1980 (JD 2444244.5)
- GPS time not adjusted by leap seconds to keep up with UT1 -> difference between GPS and UT1/UTC **increases**

$$t_{GPS} = TAI - 19s$$
$$T_{GPS} - UTC = n \cdot sec - C_0$$

- where n is integer & C_0 in order of ns
  - e.g. 2003: GPS-UTC = -13s
- GPS receiver provides access to TAI and UTC with accuracy $< 1\mu s$
- GPS phase and pseudorange measurements given in GPS time
  - GPS satellite atomic cloks fall behind $7\mu s$ mas/day w.r.t. atomic clocks on ground (due to **Special Relativity**)
- GPS satellite clocks run $45\mu s/day$ faster relative to atomic clocks on ground (**General Relativity**)
  - Net effect of $38\mu s/day$ -> position error of **10km** a day
  - In GPS: correction for compensation

DUT1: Definition

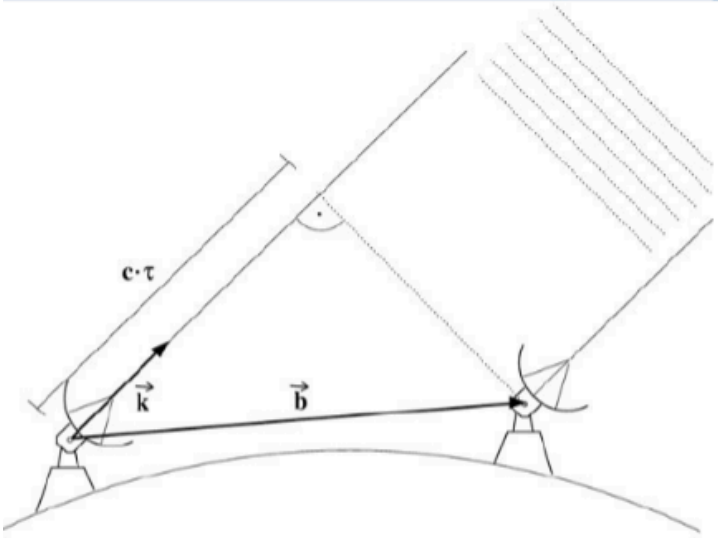
$$DUT1 = UT1 - UTC$$

- $DUT1 \in (-0.9; 0.9)$
- **leap second** to keep up with UT1 (GMT or GMSolartT corrected for polar motion)
- Values of DUT1 with precision $0.1s$ broadcasted by time signal service.
  - From DUT1-> UT1

Calculation of DUT1

- Find DUT1 by estimating UT1
  - Main Tasks of geodesy
- **VLBI** is only geodetic method allowing direct measurement of UT1
- GNSS, DORIS, SLR only determine **time derivation of UT1 (LoD)**
  - No differentiation between
    - orbital elements of satelllite ($\Omega, i, \ldots$) and
    - rotational parameters of Earth (Nutation, Precession and UT1 paramters)
  - **Orbital plans of satellite precess relative to vernal point**

UT1 Determination Principles



$$\tau = -\frac{1}{c}\vec{b}.\vec{k}$$

τ Observed time delay

$\vec{k}$ Unit vector in source direction in **CCRS**

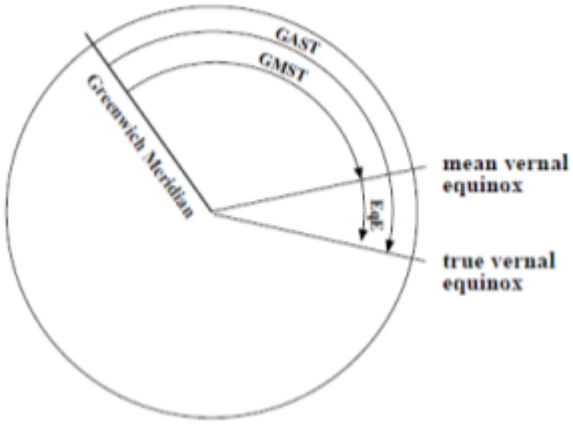
$\vec{b}$ Baseline vector in **CCTS**

- from *lecture 4* CCRS to CTRS:

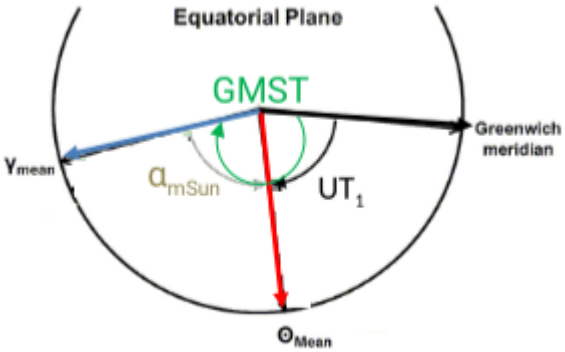
$$r_{CTRS} = S\,N\,P\,r_{CCRS}$$

$$S = R_2(-x_p)\,R_1(-y_p)\,R_3(GAST)$$

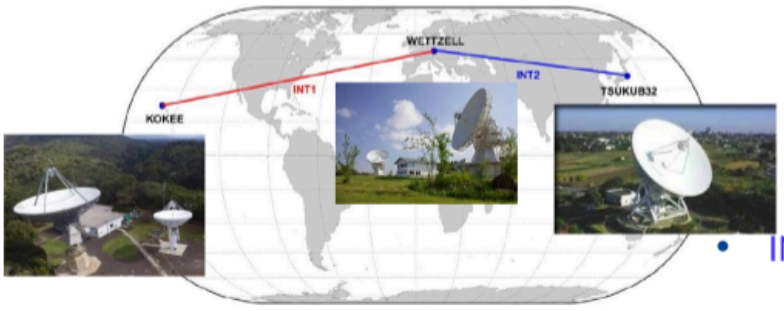
$$\tau = -\frac{1}{c}\vec{b}\boldsymbol{SNP}\,\vec{k}$$

- **Parameters** to be estimated from observations of time delays include:
  - Baseline vector (b_x, b_y, b_z)
  - Precession parameters $\boldsymbol{P}$ and Nutation elements $\boldsymbol{N}$
  - Polar coordinates x_p and y_p and **GAST** parameter
  - other parameters related to errors such as clocks, delay due to atmosphere deformation due to Earth tide, mass loading, instrumental bias, relativistic effect
- Determining GAST from VLBI τ observations:
  - GMST = GAST - $\Delta\Psi\cos(\varepsilon)$
    - Difference between mean and true vernal point due to nutation
  - 

- Given right ascension of mean sun: Determine UT1:
  - $UT1 = GMST - \alpha_{mSun}$
    - α_{mSun} is function of Julian date (IAU 1982)



- Standard VLBI for EOPs estimation based on globally distributed VLBI stations over 24 hours (multi-station network)
- **VLBI intensive (short-time) sessions**
- To determine **only** UT1: take advantage of reducing observations:
  - limit to 2 stations with a long east-west extension (single-baseline observation) over just short time period

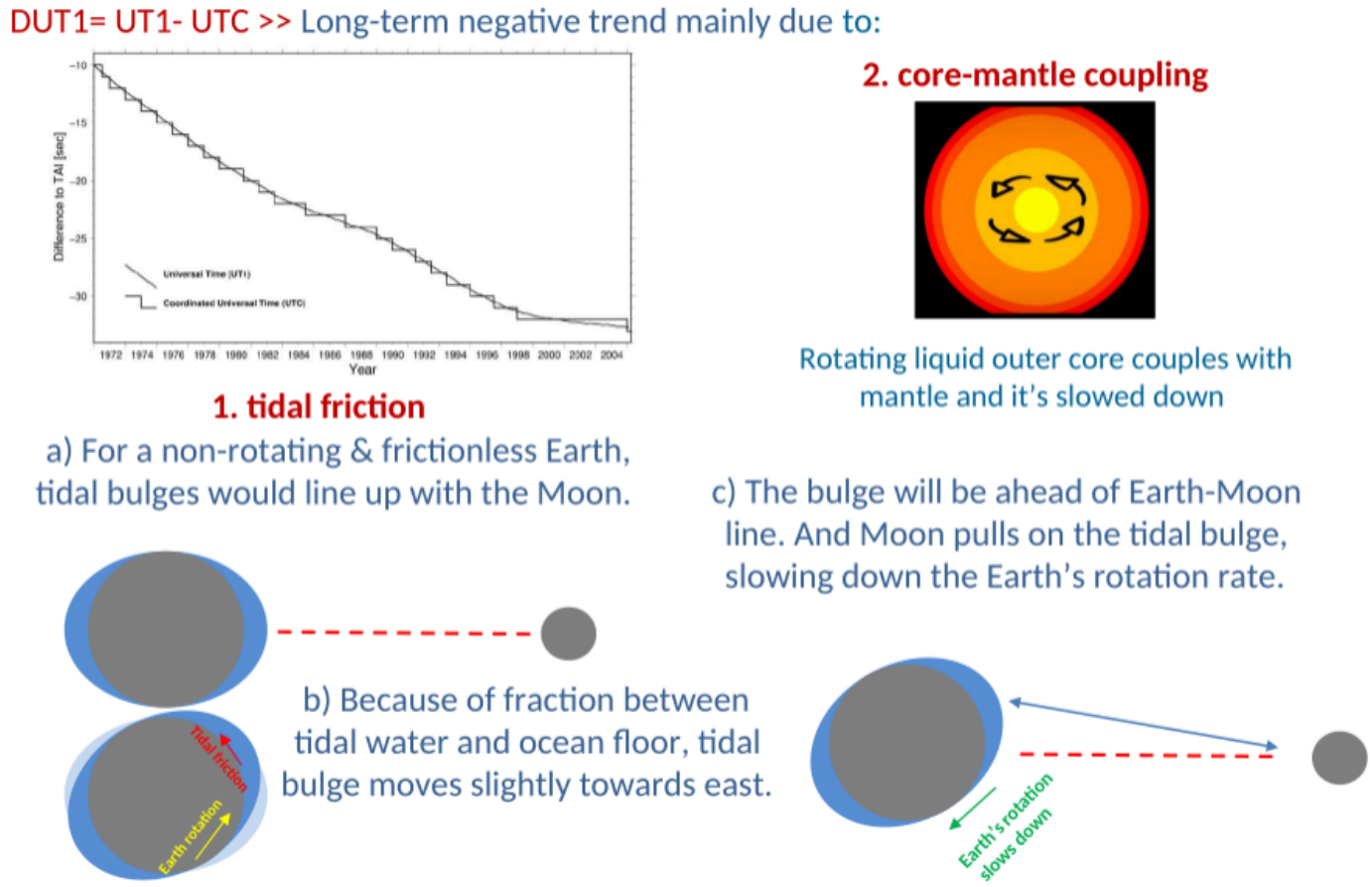


Source: Landskron & Böhm (2019)

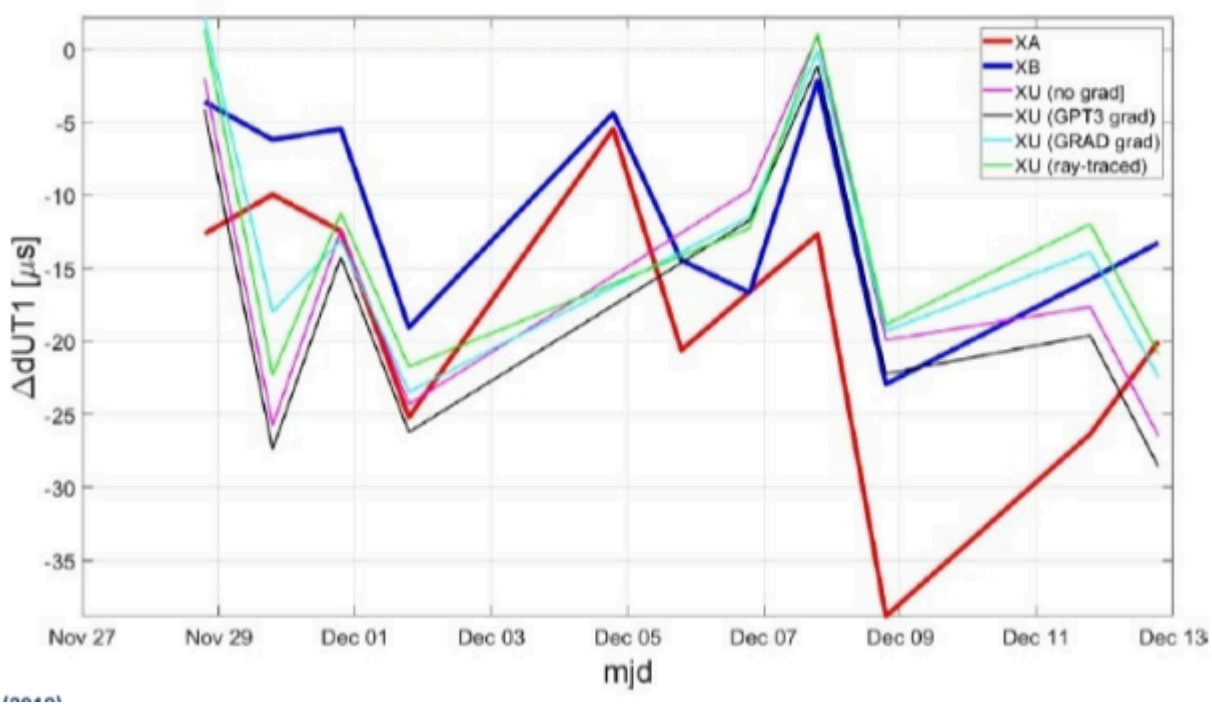
A long east-west baseline allows high sensitivity to Earth's rotation

- UT1 or DUT1 is estimated near-real-time data analysis (~ 4 mins latency)
- INT1 session: from Mon. to Fri. at 06:30 PM UTC for 1.5 hour
- INT2 session: Sat. and Sun. at 07:30 AM UTC for 1.5 hour

Long-term DUT1 Variation



Daily and Sub-Daily DUT1 Variation



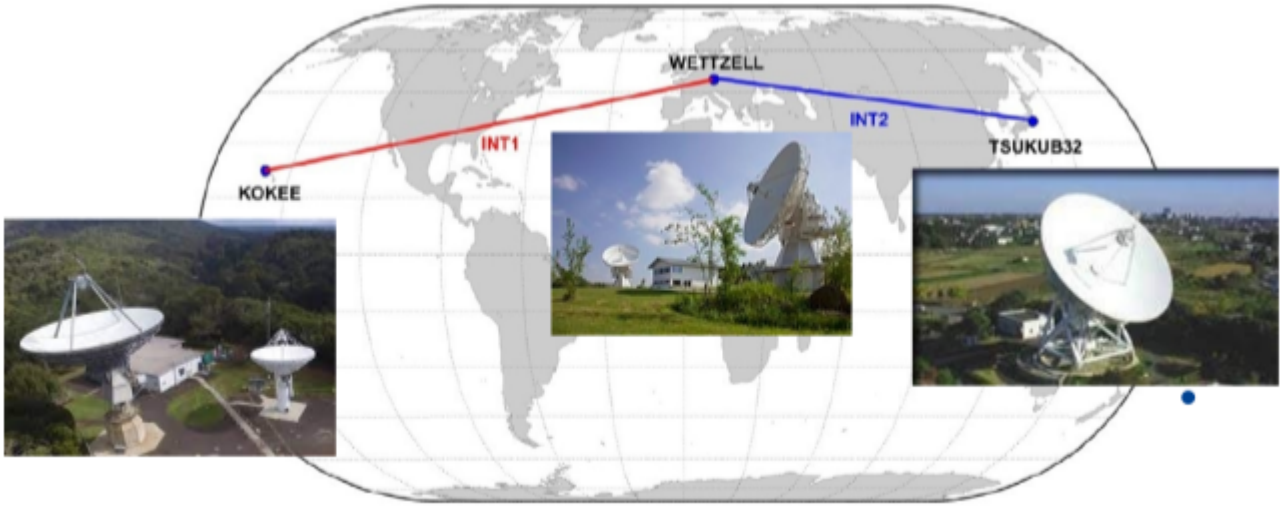
- Daily Variations of DUT1: **Length of Day (LoD)** or UT1-UTC drift
- IERS definition:
  - Difference between astronomically determined duration of the day and
  - **Nominal length** of day of 86,400 TAI seconds

- Alias **Excess on the length of day**, Earth rotation rate, LoD excitation, UT1-UTC drift or UT1 rate

- Related to instantaneous angular rate of Earth $\omega = \text{nominal rotation rate} \cdot (1 - \frac{LoD}{86{,}400}$
  - nominal rotation rate = $7.2921151467064 \times 10^{-5} \frac{rad}{s}$
  - IERS: $\omega = 72,921,151.467064–0.843,994,803 LoD$
    - ω in picoradians/s
    - LoD in miliseconds

Estimate LoD with VLBI

- LoD obtained indirectly from DUT1 values for consecutive days
- With VLBI
- $LoD = (DUT1_{t2} - DUT1_{t1})/(t2 - t1) = \Delta DUT1$

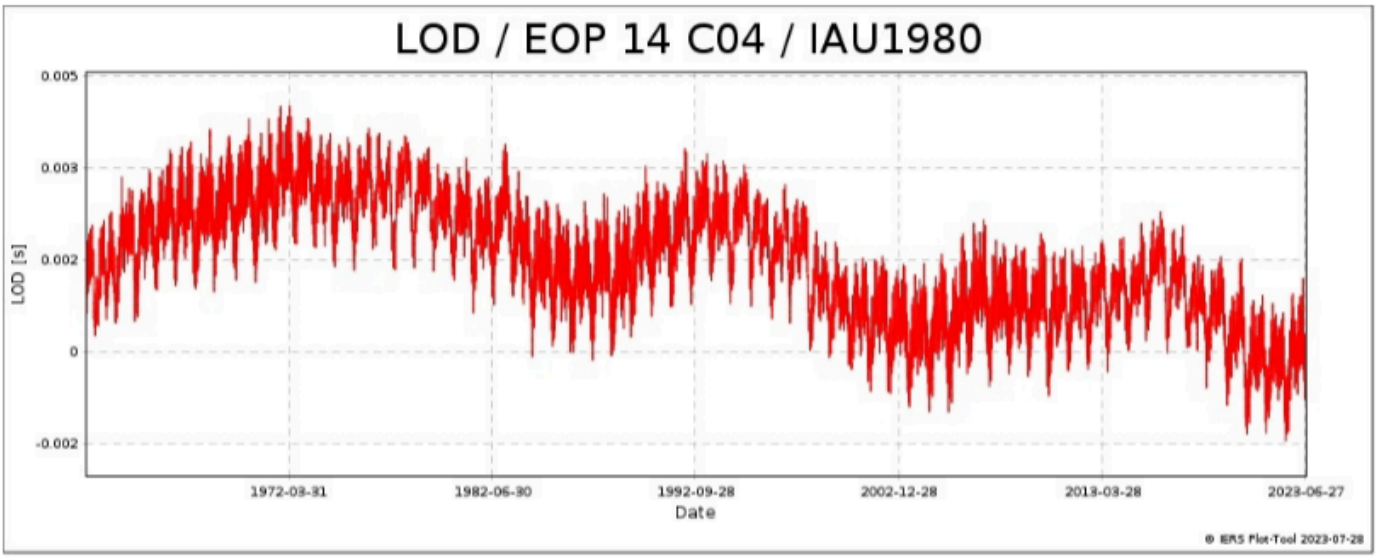
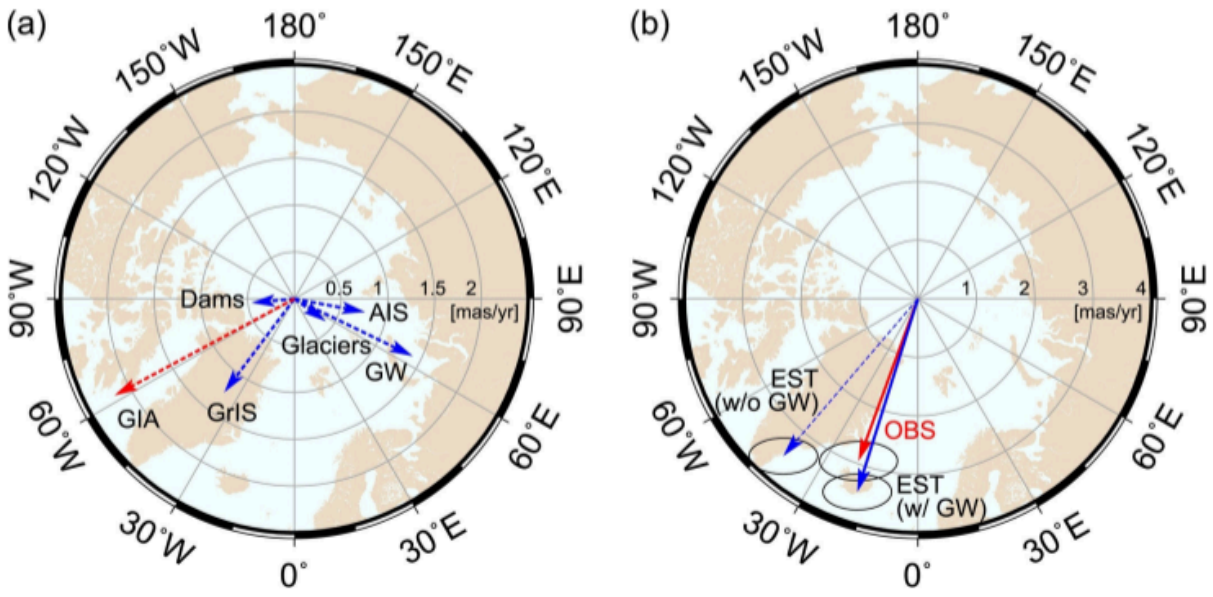


Source: Landskron & Böhm (2019)

Estimate LoD with satellite orbital analysis

- Determined from **orbital analysis** for each satellite mission
- Satellites are highly sensitive to variations in Earth's rotation, angular velocity and mass distribution
- LoD parameters derived from VLBI and combination of satellite techniques in post-processing for accuracy
- **LoD models**: Include geophysical data, atmospheric pressure data, oceanic models, hydrological models, tidal forces

Changes in LoD



- Long-term trend
  - Caused by:
    - Core-Mantle friction
    - Tidal friction
    - Glacial Rebound
- Inter-decadal variations (~8.6 years)
  - Changes due to **Geomagnetic jerks**
    - sudden changes in the Earth's geomagnetic field observed every 3-10 years
- Seasonal variation
- Effect of human
  - Pumping of 2.150 gigatons of groundwater: 6mm sea level rise from 1993-2010
  - Redistribution of water from equatorial regions to poles:
    - Accelerates Earth's rotation
    -